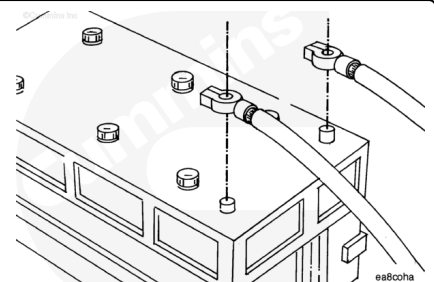


Trainings ▾

Remove

⚠ WARNING ⚠

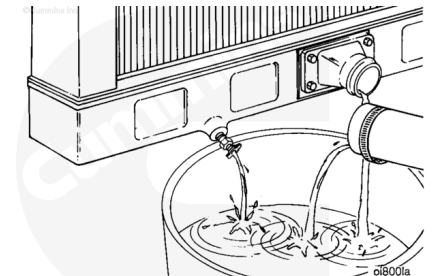
Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



Disconnect the batteries and air supply line to the air starter to prevent accidental starting. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/57/57-013-009.html>)

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

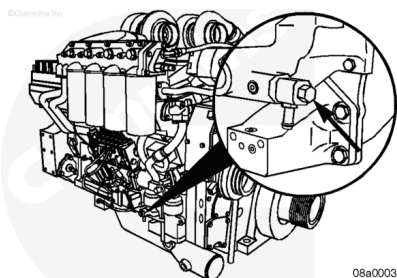


⚠ WARNING ⚠

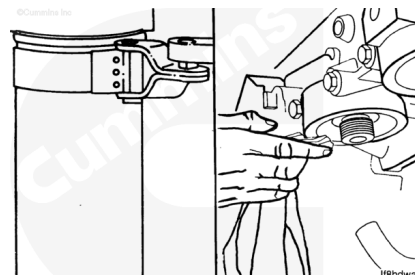
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-018.html>)

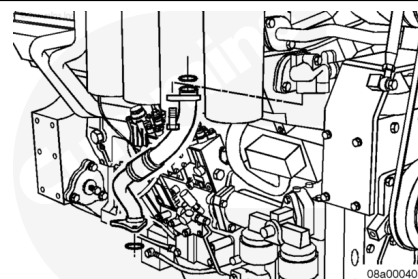
Open the draincock on the water pump outlet connection to the oil cooler. Refer to Procedure 008-062 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-062-tr.html>)



Remove the coolant filter. Refer to Procedure 008-006 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-006-tr.html>)

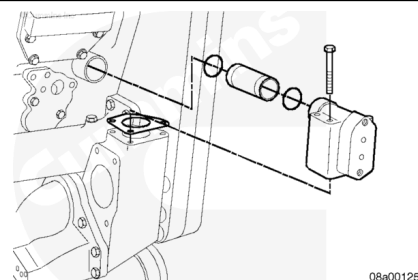


Remove the aftercooler water inlet tube and hose. Refer to Procedure 010-006 in Section 10. (</qs3/pubsys2/xml/en/procedures/57/57-010-006-tr.html>)

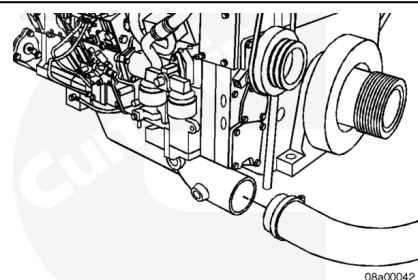


Water Bypass Connection - Remove

Remove the water bypass connection and water transfer tube.
Discard the gasket and o-ring.



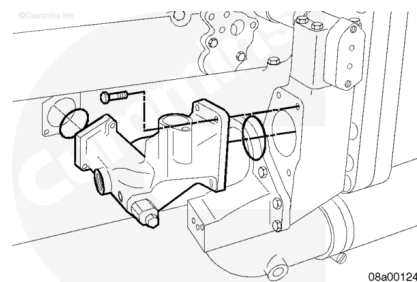
Disconnect the lower radiator hose from the water pump inlet. Refer to Procedure 008-045 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-045.html>)



Water Pump Outlet Connection - Remove

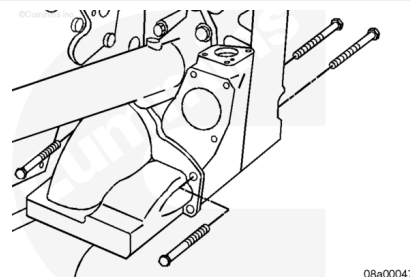
Remove the water pump outlet connection.

Discard the o-rings.



Water Pump - Remove

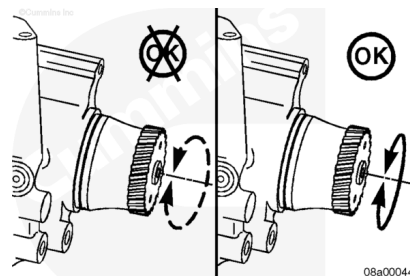
Remove the water pump.



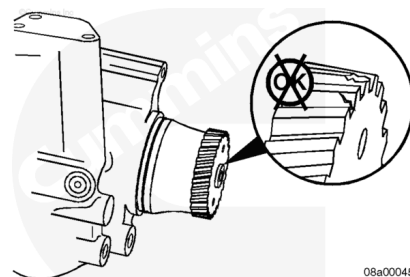
Inspect

Rotate the shaft and inspect for rough or damaged bearings.

If damage is found, replace the water pump.



Inspect the drive gear for wear; replace if worn or damaged.



Install

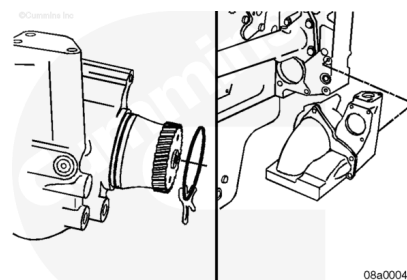
Water Pump - Install

Lubricate the o-ring with Lubriplate™ 105, and install on the water pump.

Install the water pump into the gear case.

⚠ CAUTION ⚠

Do not scratch the water pump bore with the water pump drive gear or damage the o-ring.

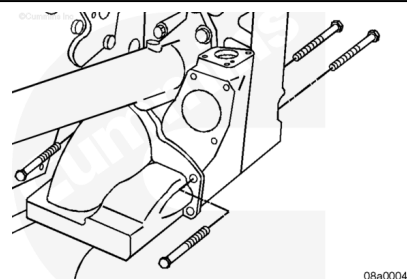


Install the four mounting capscrews.

Tighten the capscrews.

Note : Two of the capscrews install from the front of the gear cover.

Torque Value: 108 n•m [80 ft-lb]



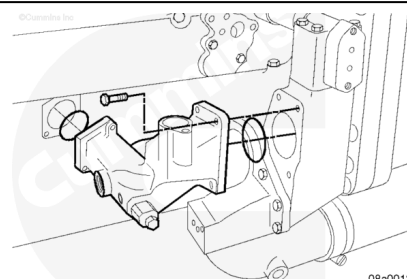
Water Pump Outlet Connection - Install

Install the o-rings in the o-ring grooves. Lubricate the o-rings with Lubriplate™ 105 grease.

Install the water pump outlet connection.

Install the capscrews and tighten.

Torque Value: 65 n•m [48 ft-lb]



Water Bypass Connection - Install

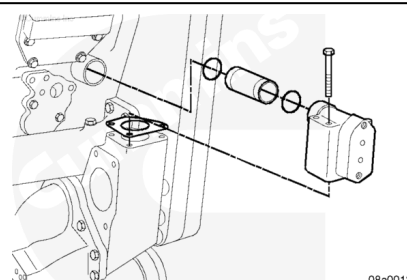
Install the water bypass connection gasket.

Install two o-rings on the water bypass connection tube. Lubricate with Lubriplate™ 105.

Insert the water bypass connection tube into the connection block first. Then assemble the connection block and tube to the engine block and water pump.

Install the capscrews.

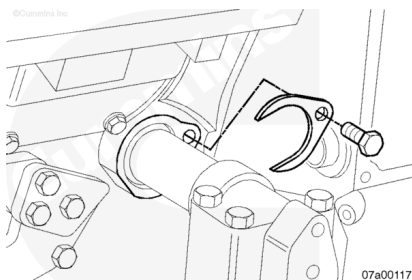
Torque Value: 65 n•m [48 ft-lb]



Install the retaining clamp.

Install the capscrew.

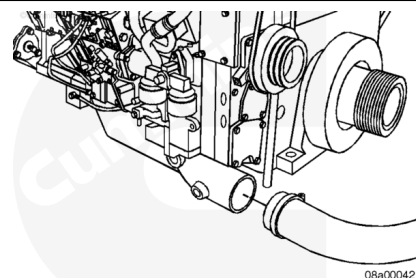
Torque Value: 65 n•m [48 ft-lb]



Connect the lower radiator hose.

Tighten the clamps.

Torque Value: 6 n•m [53 in-lb]

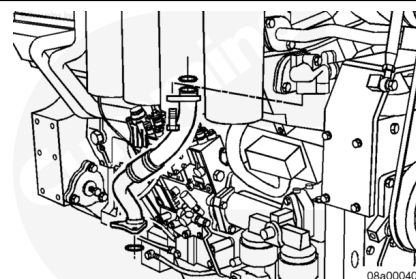


Use vegetable oil to lubricate the o-rings. Install the o-rings on both ends of the aftercooler supply tube.

Install the aftercooler supply tube and capscrews.

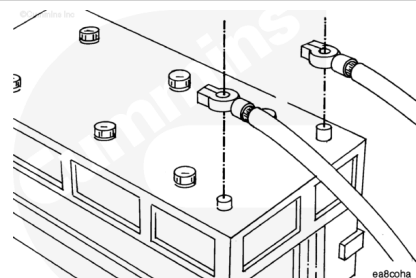
Torque Value: 65 n•m [33 ft-lb]

Note : If the tube does not align, loosen the joining hose and rotate the tubes to fit.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

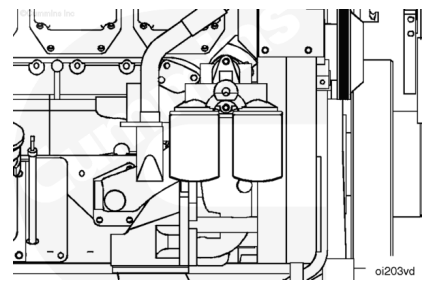


Connect the batteries. Refer to Procedure 013-007 in Section 13. (</qs3/pubsys2/xml/en/procedures/57/57-013-007.html>)

Connect the air line to the air starter. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/57/57-013-020-tr.html>)

Replace the coolant filter. Refer to Procedure 008-006 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-006-tr.html>)

Close the draincock on the water pump and oil cooler housing.
Refer to Procedure 008-062 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-062-tr.html)



Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/57/57-008-018.html)

Operate the engine to 70°C [158°F] coolant temperature and inspect for leaks.

